A SHORT-TERM INTERNSHIP REPORT ON

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Submitted in partial fulfillment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

Submitted by

Sunil Kumar (236M1A05C1)



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

B V C COLLEGE OF ENGINEERING

(Affiliated to JNTUK, Kakinada)

RAJAMAHENDRAVARAM, ANDHRA PRADESH

2026-2027

B V C COLLEGE OF ENGINEERING

(An Autonomous Institution)
(Approved by AICTE, New Delhi, Permanently affiliated to JNTUK, KAKINADA)
Palacharla, Rajamahendravaram-533102
Mobile No: 9704578666, 9948027756, E-mail: bvcr@bvcgroup.in, Website: www.bvcr.edu.in

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the internship report entitled **"ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING"** is a bonafide record of work carried out by **SUNIL KUMAR (236M1A05C1)**, B.Tech Computer Science and Engineering student at BVC College of Engineering, through SmartBridge Educational Services Pvt. Ltd., in collaboration with APSCHE, for a period of **2 months (120 hours)**, in partial fulfilment of the requirements for the award of the **Bachelor of Technology degree** from JNTU Kakinada.

Internship Coordinator
Mrs.D.Vijaya Lakshmi,
M. Tech, MISTE,(Ph.D)
Assistant Professor
Department of CSE
BVC College of Engineering.

Head of the Department
Mr.R.N.V.VishnuMurthy
M.Tech, MISTE, (Ph.D)
Associate Professor
Department of CSE
BVC College of Engineering.

EXTERNAL EXAMINER

B V C COLLEGE OF ENGINEERING

(An Autonomous Institution)

(Approved by AICTE, New Delhi, Permanently affiliated to JNTUK, KAKINADA)

Palacharla, Rajamahendravaram-533102

Mobile No: 9704578666, 9948027756, E-mail: bvcr@bvcgroup.in, Website: www.bvcr.edu.in

INSTITUTE VISION / MISSION

J.N.T.U.K CODE: 6M
SBTET CODE : 347

COUNSELLING CODE :BVCR

B V C COLLEGE OF ENGINEERING

(Approved by AICTE, New Delhi, Affiliated to JNTUK Kakinada & Govt. of A.P)
PALACHARLA(V), RAJAHMUNDARY-533102, E.G Dt. (AP)
Website: www.bvcoe.org

INSTITUTE VISION / MISSION

VISION:

To become a model adobe of learning with time trusted academic values for serving the nation and world.

MISSION:

- To provide academic infrastructure and create incubation centers.
- To augment industry – Institute Interaction through research and skill development activities.
- To build lively ambience and provide learning etiquette for all round growth of stakeholders.
- To promote innovative ideas through consultancy and knowledge hubs.
- To expand the knowledge of stakeholders by involving in workshops and training programs.

Principal

Principal
BVC COLLEGE OF ENGINEERING
PALACHARLA, RAJAHMUNDRY

B V C COLLEGE OF ENGINEERING

(An Autonomous Institution)
(Approved by AICTE, New Delhi, Permanently affiliated to JNTUK, KAKINADA)
Palacharla, Rajamahendravaram-533102
Mobile No: 9704578666, 9948027756, E-mail: bvcr@bvcgroup.in, Website: www.bvcr.edu.in

DEPARTMENT VISION, MISSION, PEOs AND PSOs DEPARTMENT VISION

BVC COLLEGE OF ENGINEERING :: PALACHARLA 533102

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

DEPARTMENT VISION

To emerge as a center to develop high quality education in the country through academic excellence and preparing the students for leadership in their fields in caring and challenging environment.

DEPARTMENT MISSION

- **M1**: To nurture high quality education with strong foundation of technologies in computer science and engineering through continuous development of infrastructure that enables the students to meet the challenges.
- **M2**: To provide an environment that values and encourages knowledge acquisition and academic freedom, making this a preferred institution for knowledge seekers.
- **M3**: In collaboration with industries, developing professionals with necessary communication skills and state-of-the-art technologies, team spirit, leadership capabilities and social responsibilities with professional ethics and human values to meet the standards.

HEAD OF THE DEPARTMENT
Head of the Department
Computer Science & Engineering
BVC College of Engineering
PALACHARLA-533 104

BVC COLLEGE OF ENGINEERING:PALACHARLA 533102
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PROGRAM EDUCATIONAL OUTCOME (PEO)

PEO1: Graduates will be in the computing profession as experts in solving hardware/software engineering problems by their depth of understanding in core computing knowledge or will be pursuing research leading to higher degrees.

PEO2: Graduates will demonstrate creativity in their engineering practices, including entrepreneurial and collaborative ventures with strategic thinking, planning, and execution.

PEO3: Graduates will communicate effectively, recognize and incorporate societal needs and constraints in their professional endeavors, and practice their profession with regard to legal and ethical responsibilities.

HEAD OF THE DEPARTMENT

BVC COLLEGE OF ENGINEERING: PALACHARLA 533102
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PROGRAM OUTCOMES

PO1. Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2. Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety, and the cultural, societal, and environmental considerations.

PO4. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6. The engineer and society: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to professional engineering practice.

PO7. Environment and sustainability: Understand the impact of professional engineering solutions in societal and environmental contexts, and demonstrate knowledge of, and need for sustainable development.

PO8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11. Project management and finance: Demonstrate knowledge and understanding of engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

ज्ञानं सर्वार्थ साधनम्

HEAD OF THE DEPARTMENT
Head of the Department
Computer Science & Engineering
BVC College of Engineering
PALACHARLA-533 102

BVC COLLEGE OF ENGINEERING: PALACHARLA 533102
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PROGRAM SPECIFIC OUTCOMES (PSO)

PSO1: Ability to apply mathematical methodologies to solve computation tasks, model real- world problems using appropriate data structures and suitable algorithms in the fields of different domains.

PSO 2: To build models and provide solutions on societal, ethical and environmental issues.

HEAD OF THE DEPARTMENT

Student's Declaration

I, SUNIL KUMAR, Reg. No. 236M1A05C1, a student of B.Tech in Computer Science and Engineering at BVC College of Engineering, hereby declare that I have successfully completed the mandatory Artificial Intelligence and Machine Learning Internship for 2 months (120 hours), organized by SmartBridge Educational Services Pvt. Ltd., in collaboration with the Andhra Pradesh State Council of Higher Education (APSCHE).

I further declare that this internship report is a genuine record of my learning, practical experience, and work carried out during the internship, submitted in partial fulfilment of the academic requirements for the award of the Bachelor of Technology degree.

Sunil Kumar
(Signature of the Student)

CERTIFICATE FROM INTERN ORGANIZATION

This is to certify that the internship report entitled "ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING" is a bonafide record of work carried out by SUNIL KUMAR (Registration Number: 236M1A05C1), a student of B.Tech in Computer Science and Engineering at **BVC College of Engineering**, who has successfully completed a two-month internship of 120 hours, from 01 June 2026 to 31 July 2026, organized by SmartBridge Educational Services Pvt. Ltd., in collaboration with the Andhra Pradesh State Council of Higher Education (APSCHE).

The report is submitted in partial fulfilment of the academic requirements for the award of the Bachelor of Technology degree.



SMARTBRIDGE

ANDHRA PRADESH STATE COUNCIL OF HIGHER EDUCATION

(A Statutory Body of the Government of A.P)

CERTIFICATE OF COMPLETION

This is to certify that Ms./Mr. **Sunil Kumar** of **Computer Science And Engineering** with Registered Hall ticket no. **236M1A05C1** under **B V C College Of Engineering** of **JNTU Kakinada** has successfully completed Short-Term Virtual Internship Program of 2 months (120 hours) on **Artificial Intelligence and Machine Learning** Organized by **SmartBridge Educational Services Pvt. Ltd.** in collaboration with **Andhra Pradesh State Council of Higher Education.**

Certificate ID: VIP-AIML-2026-4383

Date: **August 03, 2026**

Place: **Virtual**

Amarendar Katkam

Founder & CEO

Acknowledgement

I express my sincere thanks to Dr.T.V.Janardhan Rao, Principal of BVC College of Engineering, Palacharla, for helping me in many ways throughout the period of my internship with his timely suggestions.

I sincerely owe my respect and gratitude to Mr.R.N.V.Vishnu Murthy M.Tech, MISTE, (Ph.D), Associate Professor and Head of the Department of Computer Science and Engineering for his continuous and patient encouragement during all times of my internship and for helping me in completing this study successfully.

I express my special thanks to my internal coordinator, Mrs.D.Vijaya Lakshmi M.Tech, MISTE, (Ph.D), Assistant Professor, Dept. of CSE, who extended kind support in completing my internship.

I sincerely thank SmartBridge Educational Services Pvt. Ltd. and the Andhra Pradesh State Council of Higher Education (APSCHE) for providing me with the opportunity to undergo the Short-Term Internship in Artificial Intelligence and Machine Learning.

I am grateful to all the trainers and mentors for their valuable guidance, training, and feedback throughout the internship. I also extend my heartfelt thanks to everyone who directly or indirectly supported me in successfully completing this internship and report.

SUNIL KUMAR

Contents

S.NO	TOPIC	PAGE NO
1.	EXECUTIVE SUMMARY	12
2.	OVERVIEW OF THE ORGANIZATION	13
3.	INTERNSHIP PART	15
4.	ACTIVITY LOG	16
5.	OUTCOMES DESCRIPTION	22
6.	PROJECT WORK	24
7.	CONCLUSION	37
8.	REFERENCES	24

EXECUTIVE SUMMARY

The Short-Term Internship in Artificial Intelligence and Machine Learning was organized by **SmartBridge Educational Services Pvt. Ltd.**, in collaboration with the **Andhra Pradesh State Council of Higher Education (APSCHE)**. The internship was completed over a period of two months, from **01 June 2026 to 31 July 2026**, with a total duration of **120 hours**.

The internship provided practical exposure to Artificial Intelligence, Machine Learning, Deep Learning, Generative AI and Python-based technologies. The training covered fundamental AI and ML concepts, Python programming, numerical computing, data manipulation, data visualization, data preprocessing, supervised and unsupervised learning, regression, classification, neural networks and Convolutional Neural Networks.

The technical learning activities involved tools and libraries such as Anaconda Navigator, PyCharm, NumPy, Pandas, Scikit-learn, Matplotlib and Seaborn. The program also introduced application development and deployment using Python web frameworks, including Flask and FastAPI, along with frontend and database technologies.

As part of the internship, I independently developed a project entitled **"Credit Card Approval Prediction."** The project focuses on applying machine learning techniques to predict whether a credit card application is likely to be approved or rejected based on applicant information. The development work involved Python-based data processing, machine learning concepts, visualization and application development.

The primary objectives of the internship were to understand AI and ML fundamentals, gain practical programming experience, learn data preprocessing techniques, explore machine learning algorithms, understand deep learning concepts and apply the acquired knowledge through an individual project.

The internship helped strengthen my understanding of Artificial Intelligence and Machine Learning and improved my practical programming, analytical thinking, problem-solving and project development skills. It also provided an opportunity to connect academic knowledge with the practical application of machine learning in the financial domain.

OVERVIEW OF THE ORGANIZATION

A. Introduction of the Organization

SmartBridge Educational Services Pvt. Ltd. is an educational technology and skill-development organization focused on connecting academic learning with industry requirements. The organization provides industry-oriented learning programs, virtual internships, technical training and project-based learning opportunities for students.

SmartBridge works with educational institutions, industry partners and government organizations to support skill development and career readiness. Its training initiatives cover emerging technology domains such as Artificial Intelligence, Machine Learning, Generative AI, Cloud Computing, Cybersecurity, Internet of Things and other technology areas.

The organization conducted the Artificial Intelligence and Machine Learning internship in collaboration with the Andhra Pradesh State Council of Higher Education (APSCHE), providing students with structured technical learning and practical project exposure.

B. Vision, Mission and Values

Vision:

To empower learners with future-ready skills and help them develop the knowledge and competencies required for professional growth and career success.

Mission:

To bridge the gap between academic education and industry expectations through industry-academia collaboration, experiential learning and technology-driven skill development.

Core Values:

- Student-focused learning and development.
- Equal access to educational opportunities.
- Industry-oriented training and practical experience.
- Innovation and continuous improvement.
- Professionalism, excellence and integrity.
- Collaboration between educational institutions and industry.

C. Policies of the Organization in Relation to the Intern Role

The internship learning process emphasizes discipline, participation and practical skill development. The following professional practices are relevant to the internship:

- **Professionalism:** Maintaining discipline, punctuality and professional conduct.

- **Learning and Participation:** Actively participating in technical sessions and practical activities.
- **Project Completion:** Completing assigned learning activities and project work.
- **Technical Development:** Applying theoretical concepts through practical exercises.
- **Ethical Practices:** Following responsible and ethical practices while working with data and technology.

D. Organizational Structure

The broad functional structure involved in the internship program can be described as follows:

- **Management:** Provides overall organizational direction and oversees operations.
- **Program Coordinators:** Coordinate internship schedules, learning activities and program communication.
- **Technical Trainers:** Conduct technical sessions and explain concepts through demonstrations.
- **Mentors:** Guide students during practical activities and project development.
- **Interns:** Participate in training, complete learning activities and develop project work.

E. Roles and Responsibilities Related to the Internship

The internship program involved guidance and technical learning through trainers, mentors and program coordinators.

The responsibilities associated with these roles include:

- Technical trainers explain concepts, demonstrate tools and provide practical exercises.
- Mentors guide students during project development and provide feedback.
- Program coordinators communicate schedules and coordinate internship activities.
- Interns participate in sessions, practise technical concepts and complete individual or assigned project activities.

F. Performance of the Organization

SmartBridge operates in the education and technology skill-development sector. Its official website describes its work in industry-academia collaboration, virtual internships, industry-aligned courses and technology-based skilling programs.

Detailed financial information such as annual turnover, profit and market valuation has not been included in this report because verified financial figures were not available from the organizational information consulted.

G. Future Plans of the Organization

Based on its stated focus on future-ready skills and industry collaboration, the organization's areas of continued development include:

- Expanding industry-aligned learning and internship programs.
- Supporting practical and project-based technical education.
- Strengthening academic and industry partnerships.
- Promoting emerging technologies and digital skills.
- Improving career readiness through mentoring and skill development.

INTERNSHIP PART

I successfully completed a two-month Short-Term Internship in **Artificial Intelligence and Machine Learning** at SmartBridge Educational Services Pvt. Ltd., in collaboration with APSCHE, from **01 June 2026 to 31 July 2026**. The internship consisted of a total of **120 hours** of technical learning and practical activities.

The internship was designed to provide knowledge of Artificial Intelligence and Machine Learning concepts and their practical applications. The learning activities covered AI fundamentals, machine learning, deep learning, Python programming, data analysis, data preprocessing, model development and application deployment.

During the internship, I studied Python programming and libraries such as NumPy, Pandas, Matplotlib, Seaborn and Scikit-learn. I also gained exposure to TensorFlow, supervised and unsupervised learning algorithms, regression, classification, artificial neural networks and Convolutional Neural Networks.

The program additionally included Generative AI learning activities, the 100K Agentblazer Champions Program and a Generative AI course with Google Cloud.

As part of the practical learning experience, I independently developed the project **Credit Card Approval Prediction**. The project involved applying machine learning concepts to a financial prediction problem. Python-based tools and libraries were used for data processing, model development and visualization. The application development technologies included Flask, FastAPI, React, TypeScript and SQLite, as identified for the project.

The internship helped me understand the relationship between theoretical concepts and practical implementation. It also improved my programming knowledge, analytical thinking, problem-solving skills and confidence in developing AI and ML applications.

ACTIVITY LOG

WEEKLY REPORT: WEEK 1

WEEK 1 (From 01 June 2026 to 05 June 2026)

Topic: Orientation, Generative AI and AI Fundamentals

Daily Activity and Learning Outcome:

Day	Activity	Learning Outcome
Day 1	Platform orientation, internship objectives and introduction to the learning platform.	Understood the internship structure, learning process and program requirements.
Day 2	Introduction to Generative AI through a technical workshop.	Gained an introductory understanding of Generative AI concepts.
Day 3	Generative AI workshop and practical learning activities.	Explored the applications and capabilities of Generative AI.
Day 4	Introduction to the 100K Agentblazer Champions Program.	Understood the program's learning objectives and AI-related concepts.
Day 5	Continuation of the 100K Agentblazer Champions Program.	Developed familiarity with AI-driven technologies and learning resources.

Objective of the Activities: To understand the internship platform, become familiar with the program structure and gain introductory exposure to Artificial Intelligence and Generative AI.

WEEK 1 WORK REPORT:

The first week began with an orientation session introducing the internship platform, program objectives and learning activities. The initial sessions focused on Generative AI and its applications. The program also introduced the 100K Agentblazer Champions initiative, providing additional exposure to AI-related learning concepts.

Key Learnings

- Internship platform and program orientation.
- Introduction to Artificial Intelligence.
- Fundamentals of Generative AI.
- Applications of Generative AI.
- Introduction to the Agentblazer learning program.

WEEK

WEEKLY REPORT: WEEK 2

WEEK – 2 (From 08 June 2026 to 12 June 2026)

Topic: Introduction to AI, ML, Deep Learning and Python

Daily Activity and Learning Outcome:

Day	Activity	Learning Outcome
Day 1	Studied the history and evolution of Artificial Intelligence, along with the fundamental concepts of Machine Learning and Deep Learning.	Understood the development of AI and the basic differences between AI, ML and Deep Learning.
Day 2	Explored Artificial Intelligence and Machine Learning applications, real-world use cases, and commonly used tools and Python packages.	Gained awareness of practical AI/ML applications and the tools used in developing intelligent systems.
Day 3	Learned Python programming fundamentals for data science, including basic syntax, variables, data types, operators, conditional statements, loops and functions.	Developed a foundation in Python programming for data analysis and machine learning applications.
Day 4	Introduction to NumPy, Pandas, Matplotlib and Seaborn for numerical computing, data manipulation and visualization.	Understood the importance of Python libraries in handling, analyzing and visualizing datasets.
Day 5	Studied TensorFlow fundamentals, including tensors and basic data preprocessing concepts.	Gained introductory knowledge of TensorFlow and its role in developing deep learning applications.

The objective of the Activity Done: The objective of the second week was to understand the fundamentals of Artificial Intelligence, Machine Learning and Deep Learning, explore their real-world applications, and develop basic Python programming skills. The activities also introduced important Python libraries used for numerical computation, data analysis, visualization and deep learning.

WEEK 2 WORK REPORT:

The second week of the internship focused on understanding the fundamental concepts of Artificial Intelligence, Machine Learning and Deep Learning. On 08 June 2026, I studied the history and evolution of Artificial Intelligence and learned about the basic concepts and relationships between AI, ML and Deep Learning.

WEEK

Key Learnings

- History and evolution of Artificial Intelligence.
- Fundamental concepts of AI, Machine Learning and Deep Learning.
- Real-world applications and use cases of AI/ML.
- Python programming fundamentals for data science.
- Numerical computing using NumPy.
- Data manipulation and analysis using Pandas.
- Data visualization using Matplotlib and Seaborn.
- Introduction to TensorFlow and tensors.
- Basic concepts of data preprocessing.

WEEK

WEEKLY REPORT: WEEK 3

WEEK – 3 (From 15 June 2026 to 19 June 2026)

Topic: Data Preprocessing, Feature Engineering & Supervised ML Algorithms

Daily Activity and Learning Outcome:

Day	Activity	Learning Outcome
Day 1	Studied data preprocessing techniques, including handling missing values, encoding categorical data and detecting outliers.	Understood how to clean datasets and prepare data for machine learning algorithms.
Day 2	Learned feature scaling, feature selection and train-test splitting techniques.	Understood the importance of selecting relevant features and dividing datasets for model training and evaluation.
Day 3	Introduction to supervised and unsupervised machine learning, their characteristics and applications.	Learned the differences between supervised and unsupervised learning approaches.
Day 4	Studied regression algorithms, including Linear Regression, Polynomial Regression, Ridge Regression and Lasso Regression.	Gained an understanding of regression techniques and their applications in predictive analysis.
Day 5	Studied classification algorithms, including Logistic Regression, Decision Tree and Random Forest.	Understood the basic working principles of commonly used classification algorithms.

The objective of the Activity Done: The objective of the third week was to understand data preprocessing and feature engineering techniques and learn the fundamentals of machine learning. The sessions focused on preparing datasets, selecting relevant features, dividing data for training and testing, and exploring regression and classification algorithms.

WEEK 3 WORK REPORT:

The third week of the internship focused on data preprocessing, feature engineering and the fundamentals of machine learning algorithms.

WEEK

Key Learnings

- Data cleaning and preprocessing.
- Handling missing values and categorical data.
- Outlier detection techniques.
- Feature scaling and feature selection.
- Train-test splitting.
- Supervised and unsupervised machine learning.
- Linear and Polynomial Regression.
- Ridge and Lasso Regression.
- Logistic Regression.
- Decision Tree and Random Forest algorithms.

WEEK

WEEKLY REPORT: WEEK 4

WEEK – 4 (From 22 June 2026 to 26 June 2026)

Topic: Advanced ML Algorithms, TensorFlow & Neural Networks

Daily Activity and Learning Outcome:

Day	Activity	Learning Outcome
Day 1	Studied advanced classification algorithms, including Support Vector Machine (SVM), K-Nearest Neighbors (KNN) and Naive Bayes.	Understood the working principles of advanced classification algorithms and their applications in predictive modelling.
Day 2	Learned ensemble learning techniques, including Bagging, Boosting, AdaBoost and Gradient Boosting.	Gained knowledge of combining multiple models to improve prediction accuracy and model performance.
Day 3	Introduction to Artificial Neural Networks (ANN), their architecture, neurons, activation functions and layers.	Understood the basic structure and working principles of neural networks.
Day 4	Studied TensorFlow fundamentals, including tensors, computational graphs and building basic neural network models.	Learned the fundamentals of TensorFlow and its role in developing deep learning applications.
Day 5	Explored neural network training concepts, including forward propagation, backpropagation, loss functions and optimizers.	Understood how neural networks learn from training data and improve prediction performance.

The objective of the Activity Done: The objective of the fourth week was to gain knowledge of advanced machine learning algorithms, ensemble learning techniques and the fundamentals of deep learning. The sessions focused on understanding advanced classification methods, combining multiple machine learning models, exploring artificial neural network architectures and learning the basics of TensorFlow for developing neural network models.

WEEK 4 WORK REPORT:

The fourth week of the internship focused on advanced machine learning algorithms, ensemble learning, TensorFlow and neural networks.

WEEK

Key Learnings

- Support Vector Machine (SVM).
- K-Nearest Neighbors (KNN).
- Naive Bayes classification.
- Ensemble learning techniques.
- Bagging and Boosting.
- AdaBoost and Gradient Boosting.
- Artificial Neural Networks (ANN).
- Neural network architecture and activation functions.
- TensorFlow fundamentals.
- Forward propagation and backpropagation.
- Loss functions and optimizers.
- Neural network training concepts.

WEEK

WEEKLY REPORT: WEEK 5

WEEK – 5 (From 29 June 2026 to 03 July 2026)

Topic: Deep Learning, Model Evaluation & Hyperparameter Tuning

Daily Activity and Learning Outcome:

Day	Activity	Learning Outcome
Day 1	Studied deep learning fundamentals, deep neural networks and different types of neural network architectures.	Understood the concepts of deep learning and its applications in solving complex prediction problems.
Day 2	Learned model training techniques, validation methods and the importance of training, validation and testing datasets.	Gained knowledge of training machine learning models and evaluating their performance using separate datasets.
Day 3	Studied model evaluation metrics, including Accuracy, Precision, Recall, F1-Score and Confusion Matrix.	Understood how to measure classification model performance and evaluate prediction results.
Day 4	Explored hyperparameter tuning techniques, including Grid Search and Random Search.	Learned how to optimize model parameters and improve machine learning performance.
Day 5	Studied overfitting, underfitting, regularization techniques and cross-validation methods.	Understood how to improve model generalization and build reliable machine learning models.

The objective of the Activity Done: The objective of the fifth week was to understand deep learning concepts, model training procedures, evaluation techniques and hyperparameter optimization. The sessions focused on learning how to train machine learning models, measure their performance using evaluation metrics, optimize model parameters and prevent overfitting and underfitting.

WEEK 5 WORK REPORT:

The fifth week of the internship focused on deep learning fundamentals, model training, model evaluation and hyperparameter tuning.

WEEK

Key Learnings

- Deep learning fundamentals.
- Deep neural network architectures.
- Model training and validation.
- Training, validation and testing datasets.
- Accuracy, Precision and Recall.
- F1-Score and Confusion Matrix.
- Hyperparameter tuning.
- Grid Search and Random Search.
- Overfitting and underfitting.
- Regularization techniques.
- Cross-validation methods.
- Model performance optimization.

WEEK

WEEKLY REPORT: WEEK 6

WEEK – 6 (From 06 July 2026 to 10 July 2026)

Topic: ML Implementation, Dataset Analysis & Credit Card Approval Prediction Development

Daily Activity and Learning Outcome:

Day	Activity	Learning Outcome
Day 1	Explored practical implementation of machine learning algorithms using Python, Pandas and Scikit-learn.	Gained practical knowledge of implementing machine learning algorithms using Python libraries.
Day 2	Studied dataset collection, data loading, exploratory data analysis (EDA) and statistical analysis techniques.	Understood how to examine datasets, identify patterns and understand important data characteristics.
Day 3	Performed practical data cleaning, missing value handling, categorical encoding and feature preparation.	Learned how to prepare structured datasets for machine learning model development.
Day 4	Explored the implementation of classification algorithms and compared their prediction performance using evaluation metrics.	Gained practical experience in training classification models and comparing their results.
Day 5	Initiated the Credit Card Approval Prediction project by identifying project requirements, defining objectives and planning the machine learning workflow.	Understood the project development process and established a foundation for implementing the credit card approval prediction system.

The objective of the Activity Done: The objective of the sixth week was to gain practical experience in implementing machine learning algorithms and begin the development of the individual project, Credit Card Approval Prediction. The sessions focused on dataset analysis, exploratory data analysis, data cleaning, feature preparation, classification model implementation and project requirement identification.

WEEK 6 WORK REPORT:

The sixth week of the internship focused on practical machine learning implementation, dataset analysis and the initial development of my individual project, Credit Card Approval Prediction.

WEEK

Key Learnings

- Practical implementation of machine learning algorithms.
- Python libraries: Pandas and Scikit-learn.
- Dataset collection and loading.
- Exploratory Data Analysis (EDA).
- Statistical data analysis.
- Data cleaning and missing value handling.
- Categorical feature encoding.
- Feature preparation.
- Classification model implementation.
- Model performance comparison.
- Project requirement identification.
- Credit Card Approval Prediction project planning.

WEEK

WEEKLY REPORT: WEEK 7

WEEK – 7 (From 13 July 2026 to 17 July 2026)

Topic: Credit Card Approval Prediction – Data Processing, Feature Engineering & ML Development

Daily Activity and Learning Outcome:

Day	Activity	Learning Outcome
Day 1	Analyzed the Credit Card Approval Prediction dataset and examined its structure, attributes, data types and target variable.	Understood the dataset structure and identified important features required for credit card approval prediction.
Day 2	Performed data preprocessing, including handling missing values, removing duplicate records and processing categorical and numerical features.	Learned how to improve dataset quality and prepare data for machine learning model development.
Day 3	Applied feature engineering techniques, including categorical encoding, feature selection and feature scaling.	Understood how to transform raw data into meaningful features for improving model performance.
Day 4	Implemented machine learning classification algorithms using Scikit-learn and prepared training and testing datasets.	Gained practical experience in training classification models for credit card approval prediction.
Day 5	Evaluated initial machine learning models using Accuracy, Precision, Recall, F1-Score and Confusion Matrix.	Learned how to assess model performance and identify suitable classification approaches for the project.

The objective of the Activity Done: The objective of the seventh week was to process the Credit Card Approval Prediction dataset, perform feature engineering and begin developing machine learning classification models. The activities focused on understanding dataset attributes, handling missing values, transforming features, preparing training and testing datasets, implementing classification algorithms and evaluating their initial performance.

WEEK 7 WORK REPORT:

The seventh week of the internship focused on dataset processing, feature engineering and machine learning model development for my individual project, Credit Card Approval Prediction.

WEEK

Key Learnings

- Credit card dataset analysis.
- Dataset structure and attribute identification.
- Missing value handling and duplicate removal.
- Numerical and categorical data processing.
- Categorical encoding.
- Feature selection and feature scaling.
- Training and testing dataset preparation.
- Scikit-learn classification algorithms.
- Initial machine learning model development.
- Accuracy, Precision and Recall.
- F1-Score and Confusion Matrix.
- Initial model performance evaluation.

WEEK

WEEKLY REPORT: WEEK 8

WEEK – 8 (From 20 July 2026 to 24 July 2026)

Topic: Model Optimization, Evaluation & Frontend-Backend Integration

Daily Activity and Learning Outcome:

Day	Activity	Learning Outcome
Day 1	Performed hyperparameter tuning and optimization of machine learning classification models using Grid Search and Random Search.	Learned how to identify suitable model parameters and improve prediction performance.
Day 2	Compared different classification models using evaluation metrics and analyzed their prediction results.	Understood how to select an appropriate model based on classification performance and reliability.
Day 3	Developed backend functionality using Flask and FastAPI to support machine learning model integration and prediction requests.	Gained knowledge of connecting machine learning models with backend services and handling prediction requests.
Day 4	Worked on frontend development and connected user input forms with backend prediction functionality.	Understood how frontend components communicate with backend services to process user inputs and display prediction results.
Day 5	Integrated the trained machine learning model with the application and performed initial functional testing.	Learned how to combine machine learning models, backend services and frontend components into a functional application.

The objective of the Activity Done: The objective of the eighth week was to optimize machine learning models, evaluate their performance and integrate the selected prediction model into the application. The activities focused on hyperparameter tuning, model comparison, backend development using Flask and FastAPI, frontend integration and initial application testing.

WEEK 8 WORK REPORT:

The eighth week of the internship focused on machine learning model optimization, performance evaluation, backend development and frontend integration for my individual project, Credit Card Approval Prediction.

Key Learnings

- Hyperparameter tuning and model optimization.
- Grid Search and Random Search.
- Classification model comparison.
- Model performance evaluation.
- Flask backend development.
- FastAPI backend development.
- Machine learning model integration.
- Frontend and backend communication.
- User input processing.
- Prediction result generation.
- Application integration.
- Initial functional testing.

WEEK

WEEKLY REPORT: WEEK 9

WEEK – 9 **(From 27 July 2026 to 31 July 2026)**

Topic: Final Project Implementation, Testing & Documentation

Daily Activity and Learning Outcome:

Day	Activity	Learning Outcome
Day 1	Completed the integration of machine learning models with the application and reviewed the overall project workflow.	Understood the complete workflow of the Credit Card Approval Prediction system, from user input processing to prediction generation.
Day 2	Performed functional testing of the application, including input validation, backend communication and prediction functionality.	Learned how to identify functional issues and verify the correct operation of application components.
Day 3	Conducted system testing and evaluated the machine learning model using performance metrics and test data.	Gained knowledge of validating system functionality and assessing prediction model performance.
Day 4	Reviewed the application interface, corrected identified issues and organized project source code and supporting files.	Understood the importance of debugging, code organization and improving application usability.
Day 5	Completed the project documentation, summarized the development activities and reviewed the overall Credit Card Approval Prediction project.	Gained an understanding of project documentation, development reporting and the complete machine learning project lifecycle.

The objective of the Activity Done: The objective of the ninth week was to complete the Credit Card Approval Prediction project by integrating its components, performing functional and system testing, validating model performance and preparing project documentation. The activities focused on reviewing the complete application workflow, identifying and correcting issues, organizing source code and summarizing the project development process.

WEEK 9 WORK REPORT:

The ninth week of the internship focused on final project implementation, system testing, performance validation and project documentation for my individual project, Credit Card Approval Prediction.

On 27 July 2026, I completed the integration of the machine learning model with the application and reviewed the overall project workflow. I understood how the different components work together, from processing user inputs to generating credit card approval predictions.

On 28 July 2026, I performed functional testing of the application, focusing on input validation, backend communication and prediction functionality. I learned how to verify individual application features and identify possible functional issues.

On 29 July 2026, I conducted system testing and evaluated the machine learning model using test data and performance metrics. I understood the importance of validating both application functionality and model prediction performance.

On 30 July 2026, I reviewed the application interface, corrected identified issues and organized the project source code and supporting files. I also understood the importance of debugging, maintaining organized code and improving application usability.

On 31 July 2026, I completed the project documentation and summarized the major development activities carried out during the internship. I reviewed the overall Credit Card Approval Prediction project, including dataset processing, feature engineering, machine learning model development, backend implementation, frontend integration and system testing.

Overall, the ninth week marked the completion of my internship project, Credit Card Approval Prediction. It provided practical experience in developing an end-to-end machine learning application and strengthened my understanding of the complete project development lifecycle, from initial dataset analysis to final testing and documentation.

Key Learnings

- Final machine learning model integration.
- Complete application workflow.
- Functional testing and input validation.
- Backend communication testing.
- System testing and performance validation.

WEEK

- Machine learning model evaluation.
- Debugging and issue correction.
- Application interface review.
- Source code organization.
- Project documentation.
- End-to-end machine learning project development.
- Practical application of Artificial Intelligence and Machine Learning concepts.

OUTCOMES DESCRIPTION

1. Describe the Internship Environment

The internship environment provided a structured and supportive learning experience in Artificial Intelligence and Machine Learning. SmartBridge Educational Services Pvt. Ltd., in collaboration with the Andhra Pradesh State Council of Higher Education (APSCHE), organized technical learning activities that helped connect theoretical knowledge with practical applications.

The internship included technical sessions, learning modules, practical exercises and individual project development. The training activities covered Python programming, data analysis, data preprocessing, machine learning algorithms, deep learning, neural networks and application development.

The learning process encouraged independent practice, continuous improvement and the practical application of technical concepts. The available learning resources and technical guidance helped me understand different technologies and their applications in real-world problems.

Overall, the internship provided a valuable learning environment that strengthened my technical knowledge, programming abilities and confidence in developing Artificial Intelligence and Machine Learning applications.

2. Describe the Real-Time Technical Skills Acquired (In Terms of Job-Related Skills and Hands-On Experience)

During the internship, I acquired several technical skills related to Artificial Intelligence, Machine Learning, data analysis and application development.

1. Python Programming: Developed an understanding of Python programming fundamentals, including variables, data types, operators, conditional statements, loops and functions.

2. Data Analysis and Visualization: Gained practical knowledge of NumPy, Pandas, Matplotlib and Seaborn for numerical computations, dataset manipulation, statistical analysis and data visualization.

3. Data Preprocessing and Feature Engineering: Learned techniques for handling missing values, encoding categorical variables, detecting outliers, feature scaling, feature selection and train-test splitting.

4. Machine Learning Algorithms: Studied supervised and unsupervised learning approaches and explored regression and classification algorithms such as Linear Regression, Logistic Regression, Decision Tree and Random Forest.

5. Deep Learning and Neural Networks: Gained knowledge of Artificial Neural Networks, neural network architecture, activation functions, forward propagation, backpropagation and TensorFlow fundamentals.

6. Model Training and Evaluation: Learned model training procedures, performance evaluation metrics, hyperparameter tuning, cross-validation and techniques for addressing overfitting and underfitting.

7. Backend Development: Gained exposure to Python web frameworks, including Flask and FastAPI, for developing backend services and integrating machine learning functionality.

8. Frontend Development: Acquired knowledge of React and TypeScript for developing interactive user interfaces and connecting frontend components with backend services.

9. Database Management: Gained an understanding of SQLite for structured data storage and retrieval in application development.

10. Development Tools and Version Control: Used Anaconda Navigator and PyCharm for Python development and explored Git and GitHub for source code management and project organization.

11. Individual Project Development: Applied the acquired technical knowledge to independently develop the Credit Card Approval Prediction project, covering project planning, data processing, machine learning concepts and application development.

The internship activities and individual project development provided exposure to different stages of the machine learning development lifecycle, from understanding datasets to integrating prediction functionality into an application.

3. Describe the Managerial Skills Acquired

1. Project Management

- Learned to plan and organize individual project activities.
- Divided the project into different development stages.
- Established priorities for completing technical tasks.
- Followed a structured approach to project development.

2. Time Management

- Managed daily learning activities and project development tasks.
- Maintained consistency in completing weekly internship activities.
- Understood the importance of meeting project deadlines.

3. Independent Working Skills

- Developed the ability to work independently on technical assignments.
- Took responsibility for project planning, implementation and documentation.
- Improved the ability to learn new technologies through self-practice.

4. Communication and Documentation Skills

- Improved the ability to explain technical concepts.
- Developed technical documentation for project activities.
- Learned to organize project information and prepare internship reports.
- Improved the presentation of technical work.

5. Problem-Solving and Decision-Making

- Learned to identify technical problems during development.
- Explored different approaches to solving machine learning problems.
- Understood how to select suitable tools and algorithms.
- Developed analytical thinking and debugging skills.

6. Quality and Best Practices

- Understood the importance of clean and organized code.
- Learned to validate data before model development.
- Recognized the importance of model evaluation and application testing.
- Gained awareness of maintaining project files and version control.

4. Describe How You Could Improve Your Communication Skills

1. Active Participation: Participate actively in technical discussions, workshops and learning sessions to improve confidence in expressing ideas.

2. Seek and Apply Feedback: Request feedback from trainers and mentors regarding technical explanations, communication and project presentations.

3. Practise Technical Writing: Improve technical writing by preparing clear project documentation, reports, README files and code comments.

4. Improve Presentation Skills: Practise explaining project objectives, methodologies, technologies and outcomes using presentations and system architecture diagrams.

5. Practise Active Listening: Listen carefully during technical sessions, understand instructions and ask relevant questions whenever clarification is required.

6. Written Communication: Develop the ability to write clear, concise and professional emails, messages and technical reports.

7. Adapt to the Audience: Explain technical concepts using appropriate terminology for technical audiences and simpler language for non-technical audiences.

8. Continuous Learning: Participate in workshops, technical communities and hackathons to improve professional communication, confidence and knowledge sharing.

REFERENCES

The following documentation and online resources were referred to during the internship training and the development of the **Credit Card Approval Prediction** project.

1. Python Software Foundation, *Python Documentation*. Available at: https://docs.python.org/3/
2. NumPy Developers, *NumPy Documentation*. Available at: https://numpy.org/doc/
3. Pandas Development Team, *Pandas Documentation*. Available at: https://pandas.pydata.org/docs/
4. Scikit-learn Developers, *Scikit-learn User Guide*. Available at: https://scikit-learn.org/stable/user_guide.html
5. TensorFlow Team, *TensorFlow Documentation*. Available at: https://www.tensorflow.org/learn
6. Matplotlib Development Team, *Matplotlib Documentation*. Available at: https://matplotlib.org/stable/users/index
7. Seaborn Developers, *Seaborn Documentation*. Available at: https://seaborn.pydata.org/
8. Pallets Projects, *Flask Documentation*. Available at: https://flask.palletsprojects.com/
9. FastAPI, *FastAPI Documentation*. Available at: https://fastapi.tiangolo.com/
10. Meta Open Source, *React Documentation*. Available at: https://react.dev/learn
11. Microsoft, *TypeScript Documentation*. Available at: https://www.typescriptlang.org/docs/
12. SQLite Consortium, *SQLite Documentation*. Available at: https://www.sqlite.org/docs.html
13. Git Project, *Git Documentation*. Available at: https://git-scm.com/doc
14. GitHub, *GitHub Documentation*. Available at: https://docs.github.com/
15. SmartBridge Educational Services Pvt. Ltd., *Artificial Intelligence and Machine Learning Internship Learning Materials*.
16. Andhra Pradesh State Council of Higher Education (APSCHE), *Internship Programme Guidelines and Learning Resources*.

PROJECT WORK

Project Name: Credit Card Approval Prediction

Domain: Artificial Intelligence and Machine Learning

Application Type: Machine Learning-Based Classification System

Main Concept: Predicting credit card application approval outcomes using applicant-related information and machine learning techniques.

Introduction

Credit cards are widely used financial instruments that allow customers to make purchases and payments without using cash. Before issuing a credit card, financial institutions evaluate applicants based on different personal, financial and credit-related factors to determine their eligibility.

Traditional credit card approval processes may involve manual verification and assessment, which can require considerable time and effort. Machine learning provides an opportunity to analyze applicant information and identify patterns that can support the credit approval process.

The Credit Card Approval Prediction project focuses on developing a machine learning-based system that predicts whether a credit card application is likely to be approved or rejected. The system uses applicant-related information as input and applies classification techniques to generate a prediction.

The project involves several stages, including dataset analysis, data preprocessing, feature engineering, machine learning model development, training, testing and performance evaluation.

The application is designed using React and TypeScript for the frontend, Python-based backend technologies including Flask and FastAPI, and SQLite for database management. Machine learning libraries such as NumPy, Pandas and Scikit-learn support data processing and model development.

This project was developed individually as part of the Artificial Intelligence and Machine Learning internship. It provided practical experience in applying machine learning concepts to a real-world financial prediction problem.

Problem Statement

Credit card approval is an important process for financial institutions because they need to evaluate applicants before providing credit facilities. The assessment may involve analyzing multiple factors related to an applicant's personal information, income, employment and credit history.

Manual evaluation can be time-consuming and may require considerable effort when processing a large number of applications. An inefficient assessment process can also make it difficult to maintain consistency and quickly identify patterns in applicant data.

The problem is to develop a machine learning-based prediction system that analyzes applicant information, processes relevant features and predicts the possible approval outcome of a credit card application.

The proposed project aims to demonstrate how machine learning classification techniques can support preliminary credit card application assessment and provide predictions through a user-friendly application interface.

Technologies Used

Technology	**Purpose**
Python	Main programming language for machine learning and data processing.
NumPy	Numerical computations and array operations.
Pandas	Dataset loading, manipulation and analysis.
Scikit-learn	Data preprocessing, machine learning algorithms, model training and evaluation.
TensorFlow	Deep learning and neural network learning activities.
Matplotlib	Data visualization and graphical analysis.
Seaborn	Statistical data visualization.
React	Development of interactive frontend components.
TypeScript	Type-safe frontend application development.
Flask	Python backend framework included in the project technology stack.
FastAPI	Python framework included for backend API development.
SQLite	Relational database management.
Anaconda Navigator	Python environment and package management.

PyCharm	Python development and debugging environment.
Git	Version control.
GitHub	Source-code management and project repository hosting.

Features

Customer Management

Farm owners can add, update and track customer records together with the egg rate agreed with each customer (Customers & Rates module).

Egg Production Tracking

Daily egg counts are recorded shed-wise and date-wise. The dashboard shows today's production and a breakdown for each shed with its bird count and laying percentage.

Sales and Billing

A sale is created with the customer, quantity, rate and totals. The system calculates the bill amount automatically and keeps a bills ledger of all past sales.

Payments and Pending Balances

Payments are recorded against sales. The application tracks paid, unpaid and outstanding amounts for every sale and customer, and shows the number of customers with unpaid balances on the dashboard.

Feed and Nutrition

Feed purchases and usage can be added from the dashboard and viewed in the Feed & Nutrition module.

Worker Maintenance, Daily Calendar Log and Order Reminders

Tools to manage workers, keep a daily log of farm activities, and set reminders for upcoming customer orders and deliveries.

Historical Records

All records are stored permanently with date and time information, so reports are not limited to the current day.

Dashboards and Reports

Date-filtered dashboards and reports with export to Excel (SheetJS) and CSV.

Real-Time Multi-Device Sync

A sale or production entry made on one device appears on the other logged-in devices of the same business without refreshing the page.

Canvas-Based Bill Generation

Bills are drawn on an HTML5 Canvas and converted to an image that can be shared from a mobile phone using the Web Share API.

Profile Image Storage and Username Login

Profile images are stored in Supabase Storage, and users sign in with a username that is resolved to the registered email.

Architecture and Key Design Decisions

Multi-Tenancy

Each farm or business works as an isolated tenant. Every record is associated with a business_id, and access is restricted so that one business cannot view or modify the data of another business.

Data Isolation using Row Level Security

Data isolation is enforced at the database level and not only in the front-end. PostgreSQL RLS policies check the business of the authenticated user and allow read and write access only to the rows of that business. The security model therefore remains strong even if the front-end logic is bypassed.

Authentication

Login is managed by Supabase Authentication. For username-based login, the username is first resolved to its associated email through the database layer, and then the actual authentication call is made. Sessions are restored when the application is reopened.

Real-Time Synchronization

Supabase Realtime subscriptions listen for database changes. When a record such as a sale is created or updated on one device, the other logged-in devices of the same business receive the update automatically.

Billing and Bill Sharing

A sale record stores the customer, quantity, price and transaction details. Totals are calculated and the payment status is found by comparing the sale total with the recorded payments. Bills are rendered on the client side with HTML5 Canvas into an image and shared through the Web Share API on supported mobile devices, giving a proper visual invoice instead of plain text.

Pending Balance Calculation

The outstanding amount of a sale is the total sale amount minus the payments recorded for it. The pending balance of a customer is the sum of the outstanding amounts of all of their transactions.

Historical Data, Reporting and Deployment

All records are stored persistently in PostgreSQL with date and timestamp information, which supports date-filtered dashboards and reports that can be exported to Excel or CSV. The

frontend is deployed on Vercel and connected to the Git repository, so every push to the configured branch triggers an automatic build and deployment.

How System Works

1. The farm owner opens PoultryHub and logs in with the username and password.
2. The username is resolved to the registered email and Supabase Authentication creates a session for the business.
3. The dashboard loads the data of that business only, because RLS filters every table by business_id.
4. The owner records the daily egg count for each shed; the production totals and laying percentage are updated.
5. The owner adds customers with their rates and creates a sale with quantity, price and total.
6. A bill is generated on an HTML5 Canvas and shared as an image through the Web Share API.
7. Payments are recorded against the sale; the pending balance of the customer is recalculated.
8. Supabase Realtime pushes the new records to the other devices logged in to the same business.
9. The owner opens the reports, selects a date range and exports the data to Excel or CSV.
10. Reminders for upcoming orders are shown on the dashboard so that deliveries are not missed.

Folder Structure

```
PoultryHub/
├── index.html
├── package.json
├── vite.config.ts
├── tailwind.config.js
├── .env                        (Supabase URL and anon key)
└── src/
    ├── main.tsx                (application entry point)
    ├── App.tsx                 (layout, sidebar and routes)
    ├── lib/
    │   └── supabaseClient.ts
    ├── components/             (cards, forms, tables, bill canvas)
    └── pages/
        ├── Dashboard
        ├── DailyProduction
        ├── SalesBilling
        ├── CustomersRates
        ├── PaymentsLedger
        ├── FeedNutrition
        ├── WorkerMaintenance
        ├── DailyCalendarLog
        └── OrderReminders
```

Database Design

PoultryHub uses PostgreSQL through Supabase. Every business-owned table contains a business_id column, which is used by the Row Level Security policies. The main entities are:

Table / Entity	Important fields	Purpose
businesses	id, name, created_at	Stores each farm / tenant.
profiles	id, business_id, username, email, avatar_url	Links an authenticated user to a business and username.
sheds	id, business_id, name, bird_count	Stores sheds and the number of birds.
egg_production	id, business_id, shed_id, prod_date, eggs	Stores daily egg counts per shed.
customers	id, business_id, name, phone, rate	Stores customers and their egg rates.
sales	id, business_id, customer_id, sale_date, quantity, rate, total	Stores sales and bill totals.
payments	id, business_id, sale_id, customer_id, amount, paid_on	Stores payments received against sales.

Relationships: One business has many users, sheds, customers, sales and payments. One shed has many daily production records. One customer has many sales, and one sale can have many payments. The tables shown are the core entities; additional tables are used for feed, workers, calendar log and reminders.

SQL Schema

A representative PostgreSQL schema for the core tables and the data-isolation policy is shown below.

```
CREATE TABLE businesses (
  id UUID PRIMARY KEY DEFAULT
gen_random_uuid(),  name TEXT NOT NULL,
created_at TIMESTAMPTZ DEFAULT now() );


CREATE TABLE profiles (
  id UUID PRIMARY KEY REFERENCES auth.users(id),
business_id UUID NOT NULL REFERENCES businesses(id),
username TEXT UNIQUE NOT NULL,  email TEXT NOT NULL,
avatar_url TEXT );


CREATE TABLE customers (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
business_id UUID NOT NULL REFERENCES businesses(id),
name TEXT NOT NULL,  phone TEXT,  rate NUMERIC(10,2)
DEFAULT 0 );


CREATE TABLE sales (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
business_id UUID NOT NULL REFERENCES businesses(id),
customer_id UUID NOT NULL REFERENCES customers(id),
sale_date DATE NOT NULL DEFAULT current_date,
quantity INT NOT NULL,  rate NUMERIC(10,2) NOT NULL,
total NUMERIC(12,2) NOT NULL );


CREATE TABLE payments (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
business_id UUID NOT NULL REFERENCES businesses(id),
sale_id UUID NOT NULL REFERENCES sales(id),  amount
NUMERIC(12,2) NOT NULL,  paid_on DATE NOT NULL
DEFAULT current_date );
```

Back-End Implementation (Supabase)

The back-end is provided by Supabase on top of PostgreSQL, so no separate server is needed. The front-end talks to the database through the Supabase client library, and security is enforced inside the database. The core logic is described below.

1. Row Level Security Business Data Isolation

Only the rows belonging to the business of the logged-in user can be read or written.

```
ALTER TABLE sales ENABLE ROW LEVEL SECURITY;


CREATE POLICY "business_isolation" ON sales
  FOR ALL TO authenticated
  USING (business_id = (SELECT business_id FROM profiles
                        WHERE id = auth.uid()))
  WITH CHECK (business_id = (SELECT business_id FROM profiles
                             WHERE id = auth.uid()));
```

2. Supabase Client and Username-Based Login

The username is first resolved to an email through a database function, and then the normal email/password sign-in is performed.

```
import { createClient } from "@supabase/supabase-js";


export const supabase = createClient(
import.meta.env.VITE_SUPABASE_URL,
import.meta.env.VITE_SUPABASE_ANON_KEY );


export async function signInWithUsername(username: string, password: string) {
const { data: email, error } = await supabase
    .rpc("get_email_for_username", { p_username: username });
if (error || !email) throw new Error("User not found");
return supabase.auth.signInWithPassword({ email, password });
}
```

3. Pending Balance Calculation

The outstanding amount of a sale is the sale total minus its payments; the customer balance is the sum over all sales.

```
const outstanding = (sale: Sale, payments: Payment[]) =>
sale.total - payments
    .filter(p => p.sale_id === sale.id)
    .reduce((sum, p) => sum + p.amount, 0);


const customerPending = (sales: Sale[], payments: Payment[]) =>
sales.reduce((sum, s) => sum + outstanding(s, payments), 0);
```

4. Real-Time Synchronization

The page listens for changes in the sales table of the current business and refreshes the data automatically.

```
const channel = supabase   .channel("sales-
changes")
  .on("postgres_changes",
    { event: "*", schema: "public", table: "sales",
filter: `business_id=eq.${businessId}` },
    () => refreshSales())
  .subscribe();


// cleanup when the component unmounts
return () => { supabase.removeChannel(channel); };
```

5. Canvas-Based Bill Generation and Sharing

The bill is drawn on a canvas, converted to an image and shared with the Web Share API.

```
const canvas =
document.createElement("canvas"); canvas.width =
600; canvas.height = 800; const ctx =
canvas.getContext("2d")!;
ctx.fillStyle = "#ffffff"; ctx.fillRect(0, 0, 600, 800);
ctx.fillStyle = "#111827"; ctx.font = "bold 28px sans-
serif"; ctx.fillText("PoultryHub Bill", 40, 60); ctx.font =
"20px sans-serif";
ctx.fillText(`Customer: ${sale.customerName}`, 40, 120);
ctx.fillText(`Eggs: ${sale.quantity}  Rate: ₹${sale.rate}`, 40, 160);
ctx.fillText(`Total: ₹${sale.total}`, 40, 200);


canvas.toBlob(async (blob) => {
  const file = new File([blob!], "bill.png", { type: "image/png" });
if (navigator.canShare?.({ files: [file] })) {     await
navigator.share({ files: [file], title: "PoultryHub Bill" });
  }
}, "image/png");
```

6. Excel Report Export

```
import * as XLSX from "xlsx";


const ws = XLSX.utils.json_to_sheet(rows); const
wb = XLSX.utils.book_new();
XLSX.utils.book_append_sheet(wb, ws, "Sales");
XLSX.writeFile(wb, "sales-report.xlsx");
```

Front-End Code

The front-end is developed with React, TypeScript, Vite and Tailwind CSS. The application is a single-page application: the browser loads index.html, which contains an empty root element and a module script that starts the React application from /src/main.tsx. All pages, the sidebar and the dashboard cards are rendered by React components styled with Tailwind CSS utility classes.

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=1.0, user-scalable=no, viewpo
    <meta name="theme-color" content="#059669" />
    <meta name="apple-mobile-web-app-capable" content="yes" />
    <meta name="apple-mobile-web-app-status-bar-style" content="default" />
    <meta name="application-name" content="PoultryHub" />
    <meta name="apple-mobile-web-app-title" content="PoultryHub" />
    <title>PoultryHub</title>
    <meta name="description" content="Poultry Farm Business Management mobile application for tracking daily egg prod
    <meta property="og:title" content="PoultryHub" />
    <meta property="og:description" content="Poultry Farm Business Management mobile application for tracking daily e
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link href="https://fonts.googleapis.com/css2?family=Plus+Jakarta+Sans:wght@400;500;600;700;800&family=Outfit:wght
  </head>
  <body class="bg-[#F9FAFB] text-[#111827] font-sans antialiased selection:bg-[#D1FAE5] selection:text-[#065F46]">
    <div id="root"></div>
    <script type="module" src="/src/main.tsx"></script>
  </body>
</html>
```

Figure 1: entry file of the PoultryHub front-end

- apple-mobile-web-app meta tags make the application behave like a mobile app on phones (name “PoultryHub”).
- The title, description and Open Graph tags describe the application as a Poultry Farm Business Management mobile application for tracking daily egg production, sales and payments.
- Google Fonts (Plus Jakarta Sans and Outfit) are preconnected and loaded for the interface typography.
- The body uses Tailwind utility classes for the background colour, text colour, font, antialiasing and the green text-selection colour.
- The div with id “root” is the mount point, and the script of type “module” loads /src/main.tsx, which starts the React application through Vite.

2. User Interface of the Application

Figure 2 shows the dashboard of the running application for a sample farm account. The interface is organized as follows:

- **Left sidebar:** the PoultryHub logo and farm name, the cloud-sync status with a “Sync Farm Data” button, and the navigation grouped as Core Operations (Dashboard, Daily Production, Sales & Billing, Customers & Rates, Payments & Ledger, Feed & Nutrition) and Tools & Management (Worker Maintenance, Daily Calendar Log, Order Reminders). The logged-in farm owner and the logout button are at the bottom.

- **Summary cards:** Today's Production (eggs), Total Sales Today (in rupees), Pending Payments (number of customers unpaid) and Total Flock (number of birds and active sheds, with a Manage link).
- **Quick actions:** one-tap buttons for Egg Count, Add Sale, Customer, Add Feed and Reminder.
- **Recent Sales Activity:** today's transactions in newest-first order, with buttons to create a new bill or open the Bills Ledger.
- **Production Breakdown:** shed-wise bird count, eggs (with trays) and laying percentage, with an Add Shed option.
- **Upcoming Orders:** delivery alerts for the day, with a Manage link.

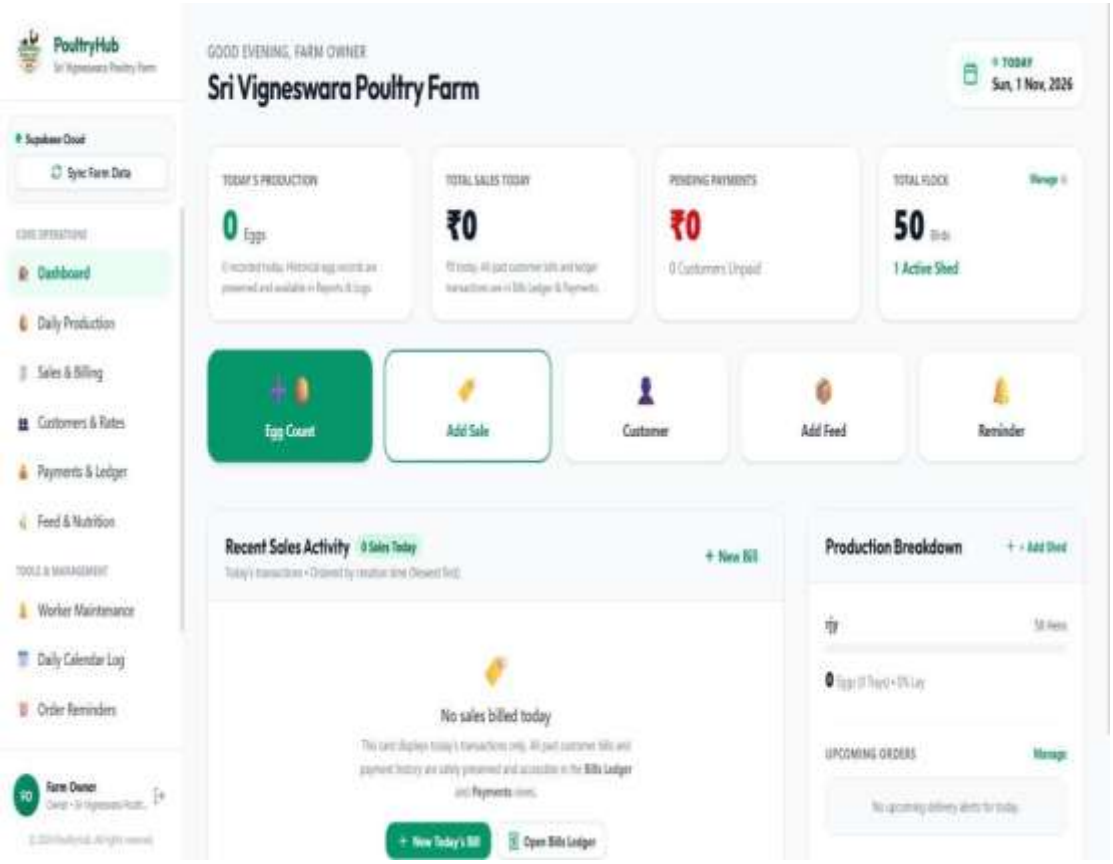


Figure 2: PoultryHub Dashboard

How to Run

- Install Node.js 18 or later and Git, and create a free Supabase project.
- In the Supabase SQL editor, create the tables and enable Row Level Security with the business-isolation policies.
- Clone the project and create a .env file with VITE_SUPABASE_URL and VITE_SUPABASE_ANON_KEY.
- Open a terminal in the project folder and run: npm install.
- Run: npm run dev, then open the local Vite URL shown in the terminal, normally http://localhost:5173.

- Log in, add a shed, customers and rates, then record the egg count and create a sale.
- Generate and share a bill, record a payment and check the pending balance on the dashboard.
- Open the application on a second device with the same login to observe real-time synchronization.
- To deploy, push the code to GitHub and import the repository in Vercel; every push triggers a new deployment.

Important Modules

Module	Purpose
Dashboard	Summary of production, sales, pending payments, flock, recent sales and upcoming orders.
Daily Production	Record and view daily egg counts for each shed.
Sales & Billing	Create sales, generate bills and view the Bills Ledger.
Customers & Rates	Manage customers and their egg rates.
Payments & Ledger	Record payments and track outstanding balances per customer.
Feed & Nutrition	Record feed purchases and consumption.
Worker Maintenance	Manage farm workers.
Daily Calendar Log	Daily log of farm activities.
Order Reminders	Reminders for upcoming customer orders and deliveries.

Testing

The PoultryHub application will be tested to ensure that all major features work correctly and provide a smooth experience for the farm owner. Testing begins with username-based login, session restoration after reopening the application, and logout. It will verify that each user sees only the data of their own business.

The production and sales modules will be tested by adding sheds, daily egg counts, customers and sales, and checking that the dashboard cards, bill totals and shed-wise laying percentages are calculated correctly. The payment module will be tested with full, partial and multiple payments to confirm that the outstanding amount of each sale and the pending balance of each customer are correct.

Data isolation will be tested by logging in with two different businesses and confirming that neither can read or modify the records of the other. Real-time synchronization will be tested by opening the application on two devices and checking that a new sale appears on the second device without a manual refresh.

Finally, bill generation and sharing will be tested on a mobile browser, report filtering by date range will be tested with Excel and CSV export, and the responsive layout will be checked on different screen sizes. These testing activities help identify errors, improve accuracy and ensure that PoultryHub provides reliable farm information.

Project Conclusion

PoultryHub provides a centralized and secure solution for managing poultry farm operations. A farm can record daily production, manage customers, create sales, generate and share digital bills, track payments and pending balances, and view historical data and reports in one application. By combining multi-tenant PostgreSQL security, real-time multi-device synchronization and Canvas-based image billing, the project goes beyond a basic CRUD application and gives the business better visibility into its production, sales and payments.

Future Scope

- Improve offline reliability so that entries can be made without internet and synchronized later.
- Add stronger monitoring and error tracking.
- Optimize database queries as the data volume grows.
- Introduce advanced analytics and forecasting for production and sales.
- Send automatic payment reminders to customers with pending balances.
- Add multi-language support for farm owners and workers.

CONCLUSION

The short-term internship on Web Development at OTP Technologies Pvt. Ltd., Eluru, in collaboration with the Andhra Pradesh State Council of Higher Education (APSCHE), was a valuable learning experience that provided practical knowledge of modern web development technologies and their real-world applications. Throughout the 120-hour internship, I gained an understanding of the complete web development process, including designing user interfaces, developing interactive web applications, designing databases, securing data and integrating the front-end with back-end services.

During the internship, I learned and practised various technologies, including HTML5, CSS3, JavaScript, TypeScript, React, Vite, Tailwind CSS, Supabase and PostgreSQL. I also gained experience using Visual Studio Code, Git, GitHub and Vercel. The weekly assignments and practical activities helped me strengthen my programming knowledge, understand application development concepts and improve my ability to identify and solve technical problems.

As part of the project work, I developed PoultryHub – a multi-tenant farm management and billing web application. The application manages egg production, customers, sales, bills, payments, pending balances and reports, uses PostgreSQL Row Level Security for businesslevel data isolation, synchronizes data across devices in real time and generates shareable image bills. It demonstrates how modern front-end and cloud back-end technologies can be integrated to solve a practical problem of poultry farm management.

Overall, this internship successfully bridged the gap between academic learning and practical software development. It provided a strong foundation in full-stack web development and enhanced my confidence to learn new technologies, work on real-world applications and take up future challenges in the software industry. The knowledge and experience gained through this internship will be valuable for my academic growth and future professional career.